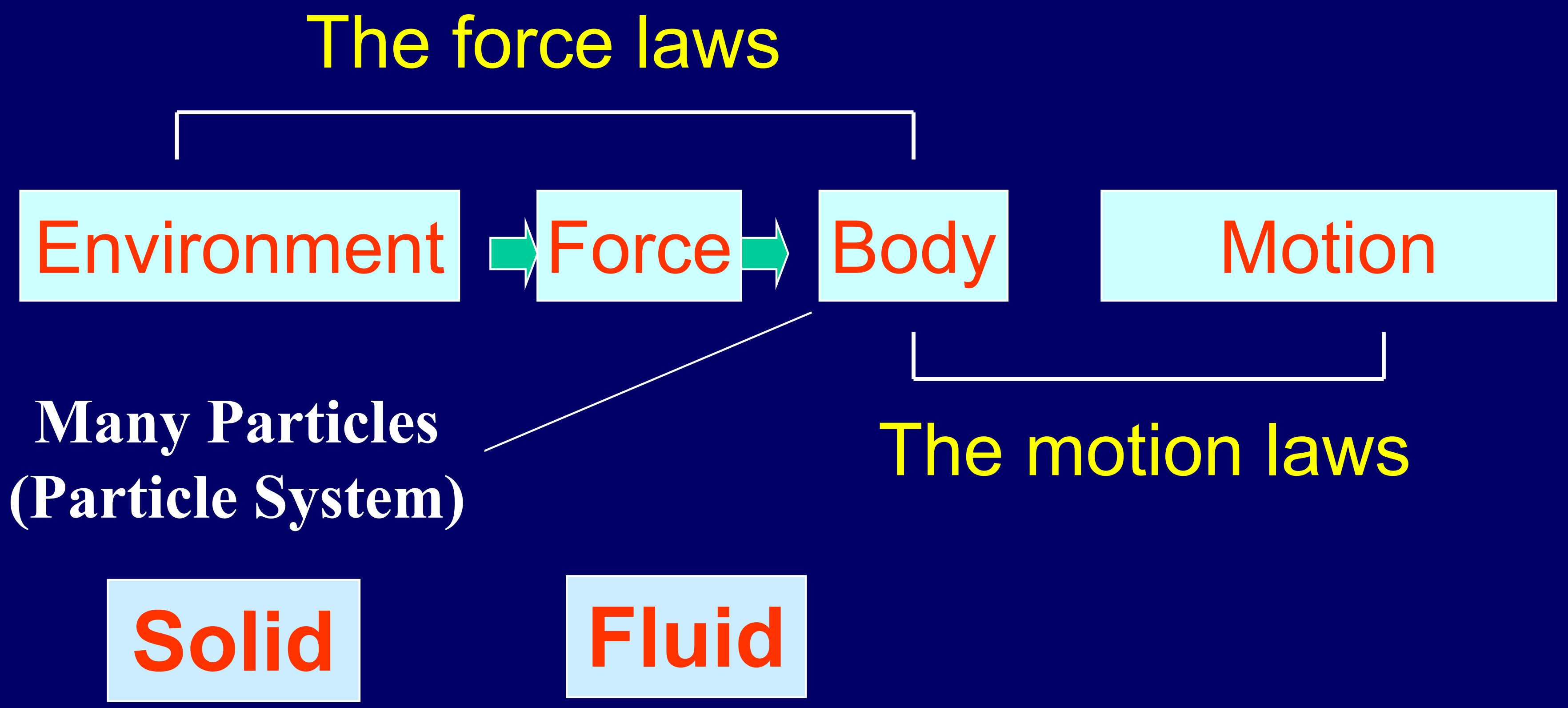
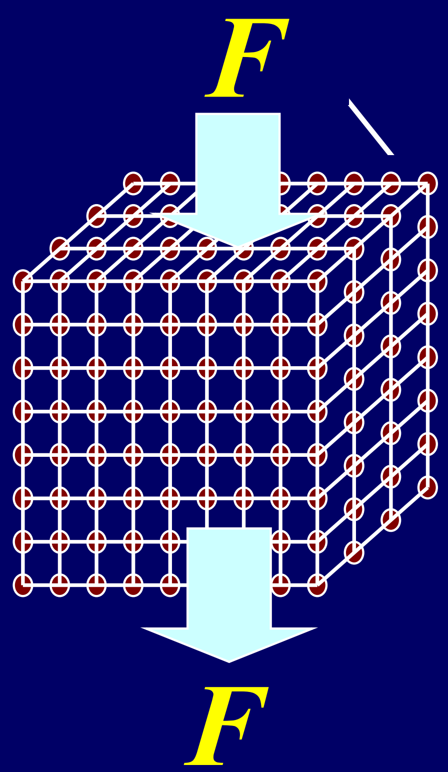
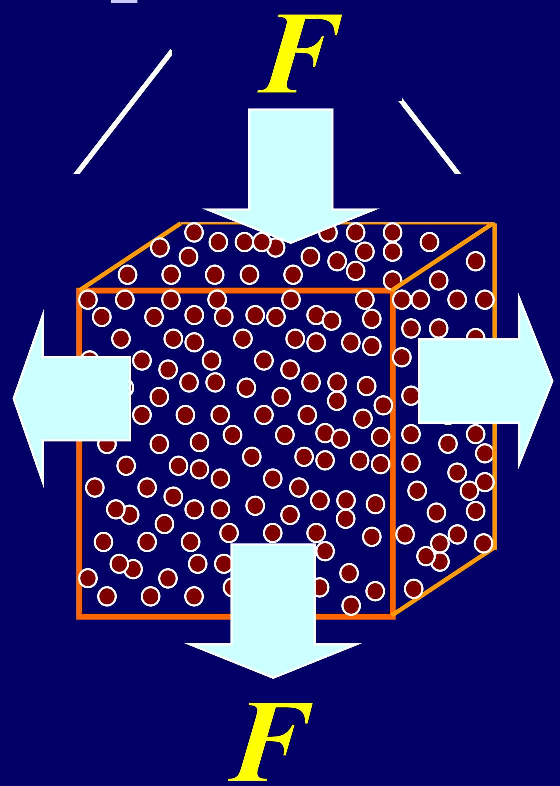

The force laws
Environment
Force
Body
Motion
Many Particles
(Particle System)
The motion laws
Solid
Fluid

CHAP.15
FLUID STATICS

Many particles matter

Solid Liquid Gas





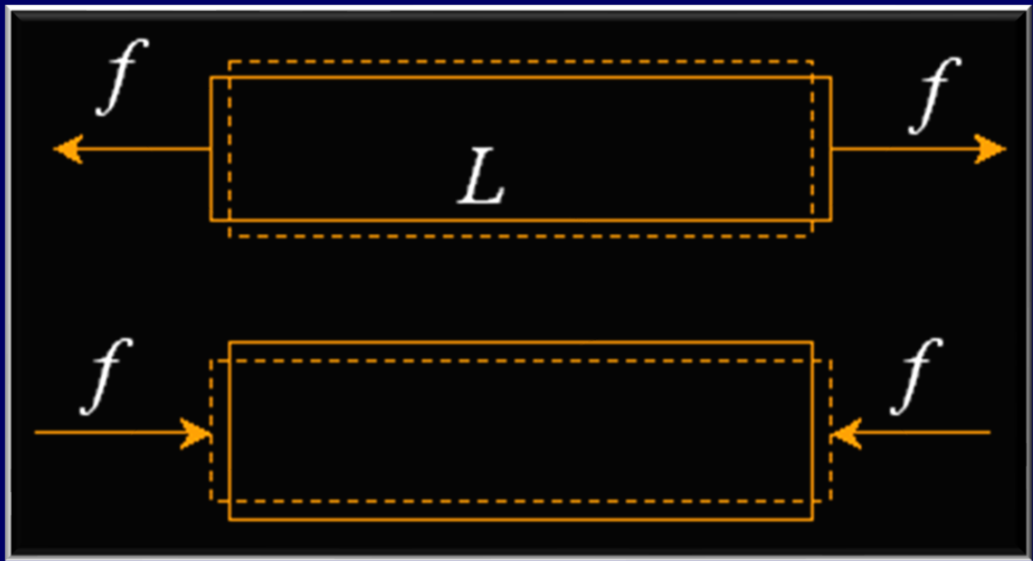
$$F = ?$$

$$p = \frac{\Delta F}{\Delta A}$$

Pressure

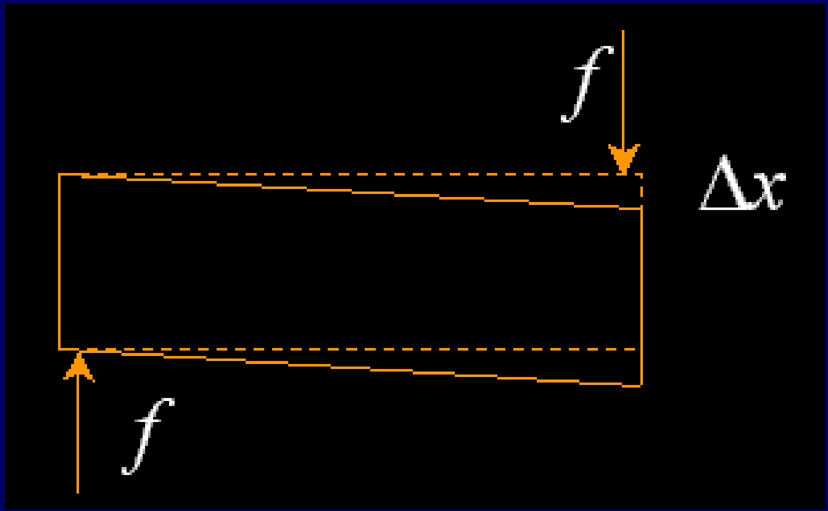
Fluids and Solids

solids: stress and strain (应力和应变）



$$\frac{f}{A} = Y\frac{\Delta L}{L}$$

Y: Young’s modulus



Shearing stress

$$\frac{f}{A} = G\frac{\Delta x}{L}$$

G: Shear modulus

solid: compression force, tensile force, shearing force

liquid: compression force, tensile force, no shearing force

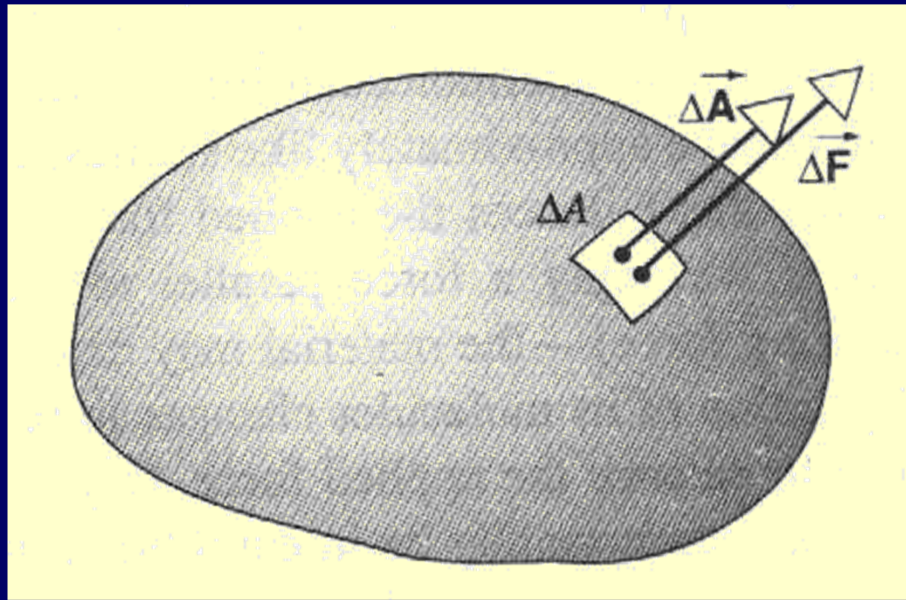
gas: compression force, no tensile force, no shearing force

Pressure and Density

Pressure

$$\Delta\vec{F} = p \cdot \Delta\vec{A}$$

$$\boxed{p = \frac{\Delta F}{\Delta A}}$$



The unit of pressure: pascal (1Pa= 1N/m^2), or atmosphere (atm), bar (1bar=10^{5}Pa), mm of Hg.

Density:

$$\rho = \frac{\Delta m}{\Delta V} \quad \Rightarrow \quad \rho = \frac{m}{V} \text{ or } \rho = \frac{dm}{dV}$$

Hydraulic stress

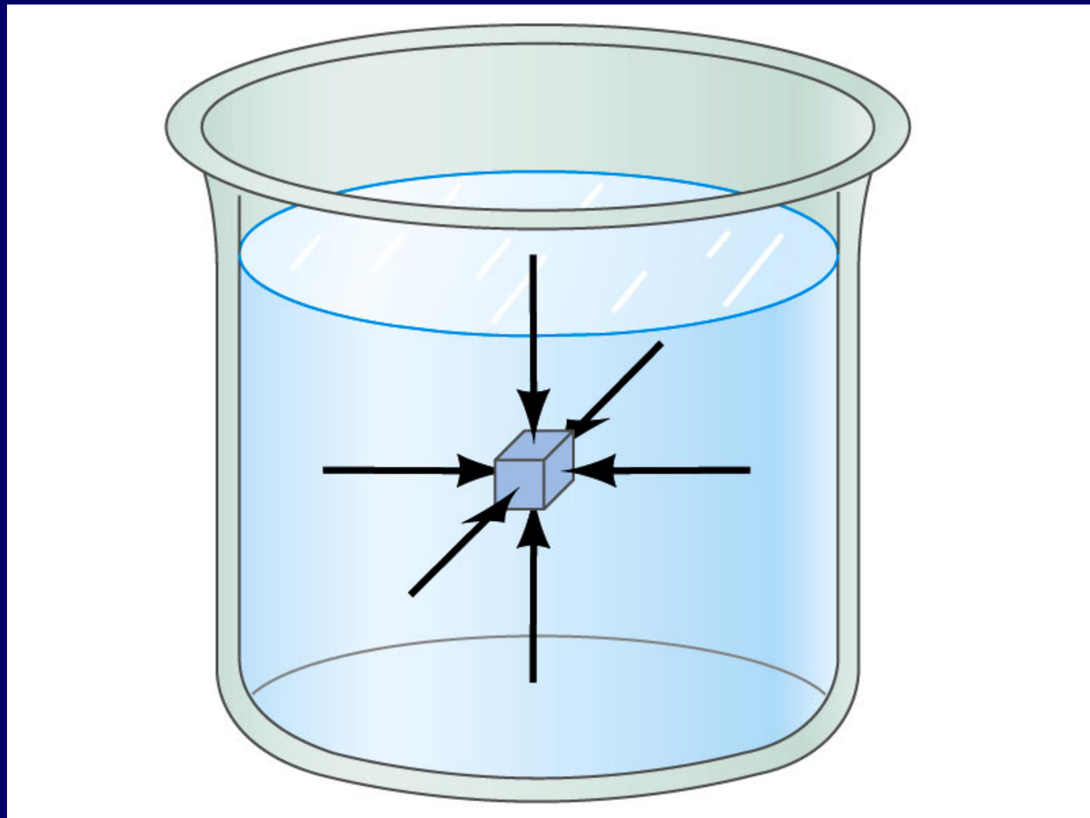
Bulk modulus:

$$B = -\frac{\Delta p}{\Delta V / V}$$

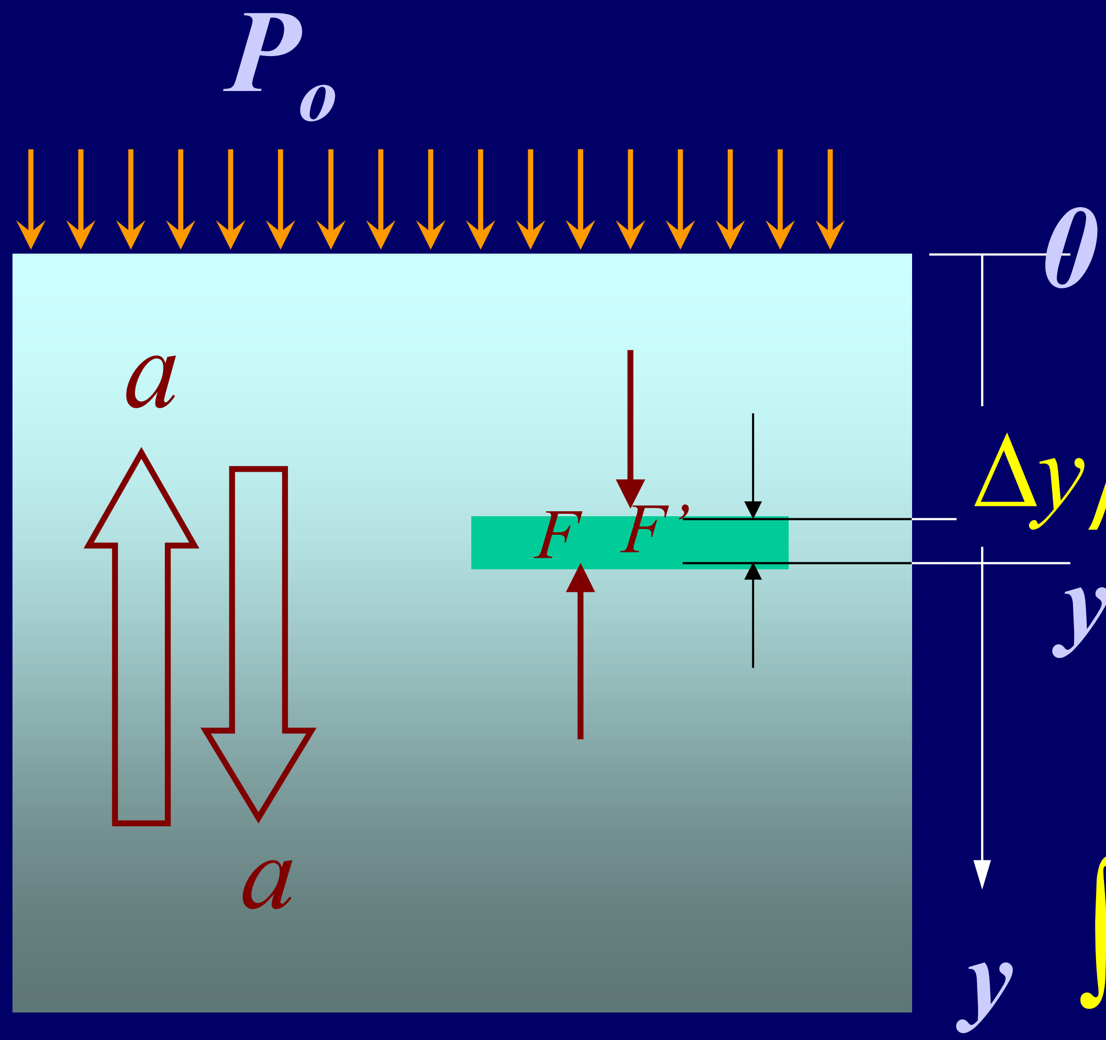
$$\Delta \rho = -\frac{m}{V^2} \Delta V = \frac{m}{BV} \Delta p = \frac{\rho}{B} \Delta p$$

Fluids at Rest

Pressure is the same in every direction in a static fluid at a given depth; if it were not, the fluid would flow.



Pressure in the liquid



$$\Delta F = F - F'$$

$$= Ap_{y+\Delta y} - Ap_y$$

$$= A\Delta y \rho g + A\Delta y \rho a$$

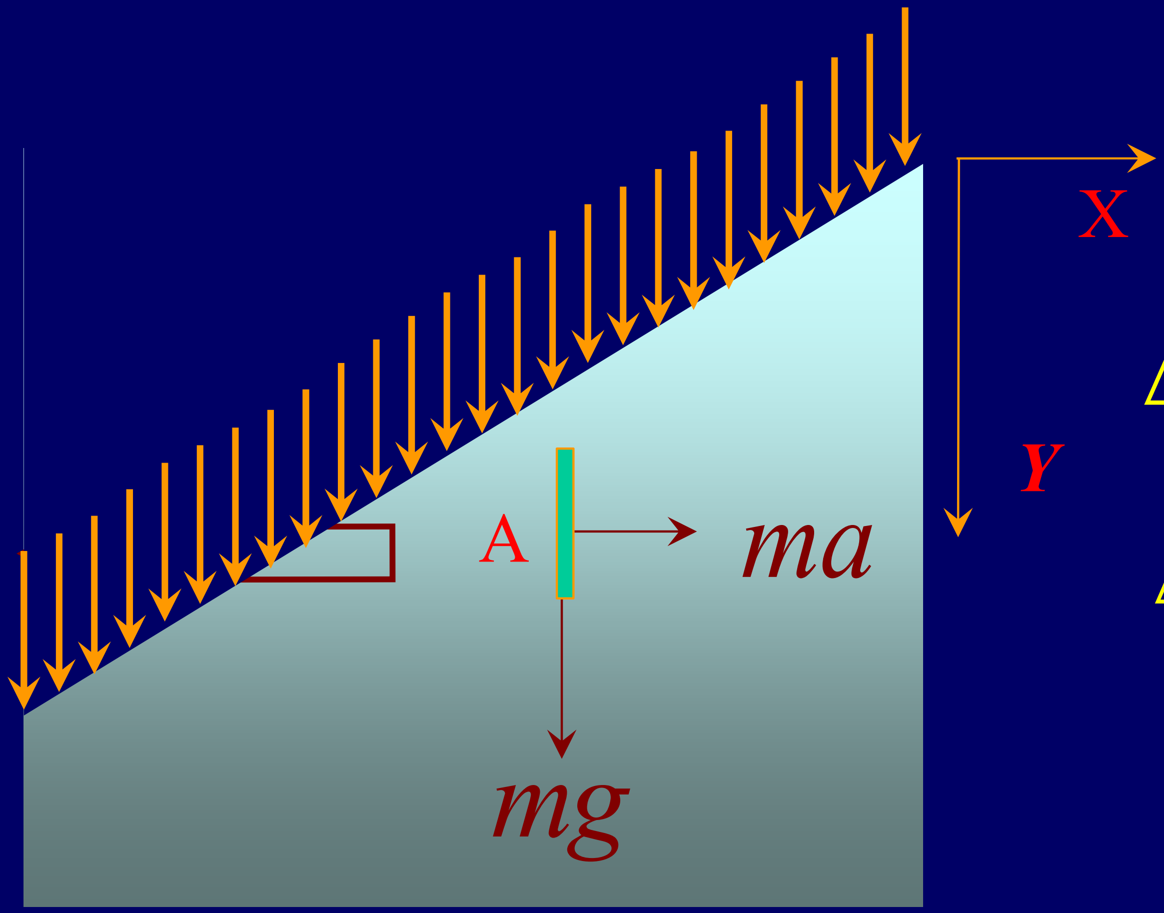
$$\Delta y \rho (g + a) = p_{y+\Delta y} - p_y$$

$$\rho (g + a) dy = dp$$

$$\int_0^y \rho (g + a) dy = \int_{p_0}^{p} dp$$

$$p = p_0 + \rho (g - a) y$$

Pressure in the liquid



$$\Delta F = F - F'$$

$$= Ap_{x+\Delta x} - Ap_x$$

$$= \Delta x A \rho a$$

$$\Delta x \rho a = p_{x+\Delta x} - p_x$$

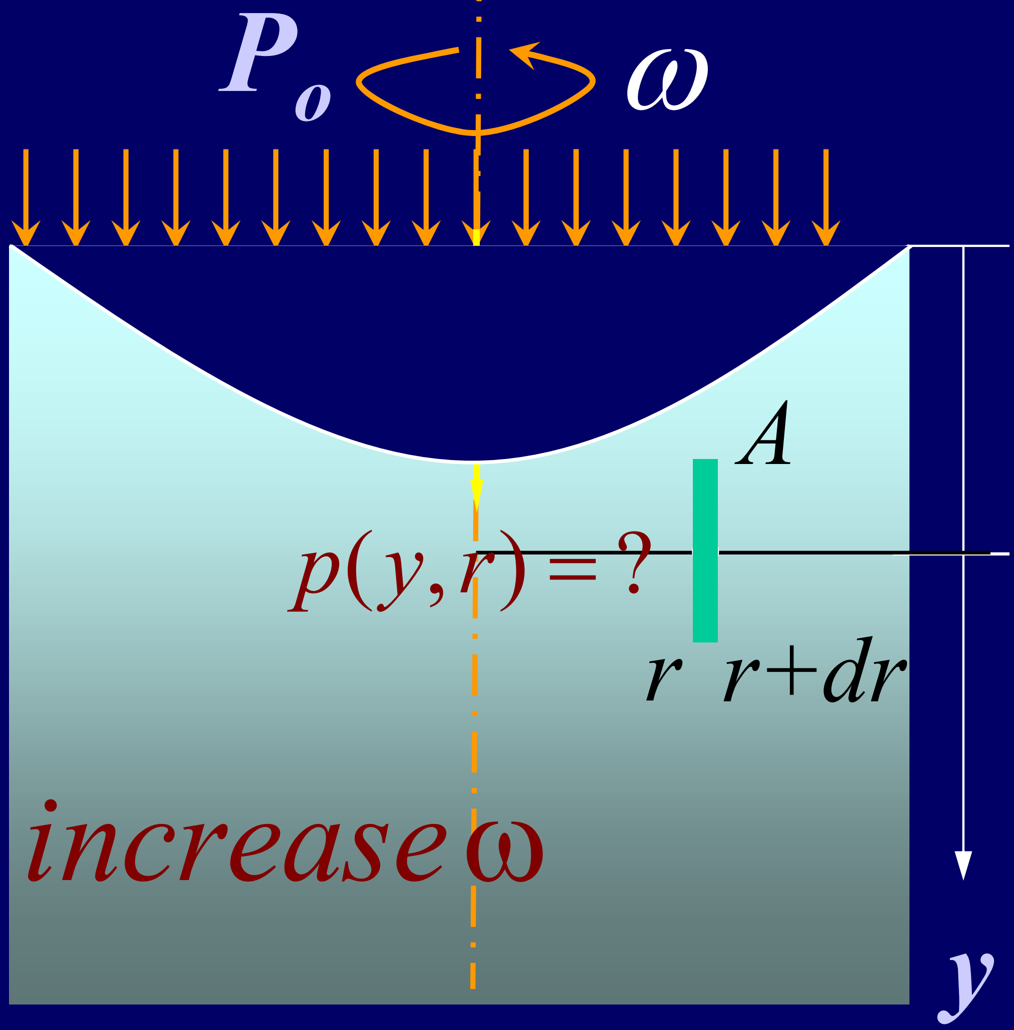
$$\Delta x \to 0 \quad \rho a dx = dp$$

$$\int_0^x \rho a dx = \int_{p_0}^{p} dp$$

$$p(y,x) = p(y,0) + \rho a x$$

$$p(x,y) = p_0 + \rho a x + \rho g y \quad y = -ax/g$$

Pressure in the liquid



On the surface

$$p_0 = p_0 + \rho g y + \frac{\rho\omega^2 r^2}{2}$$

$$\rho g y + \frac{\rho\omega^2 r^2}{2} = 0$$

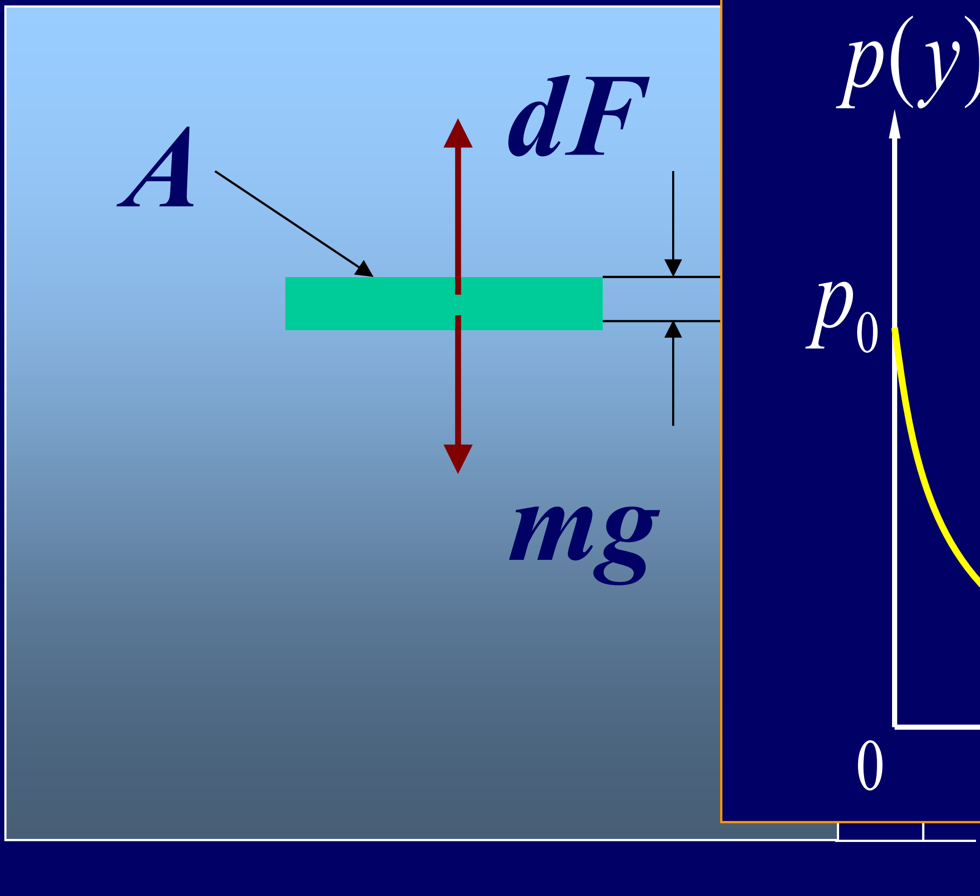
$$y = -\frac{\omega^2 r^2}{2g}$$

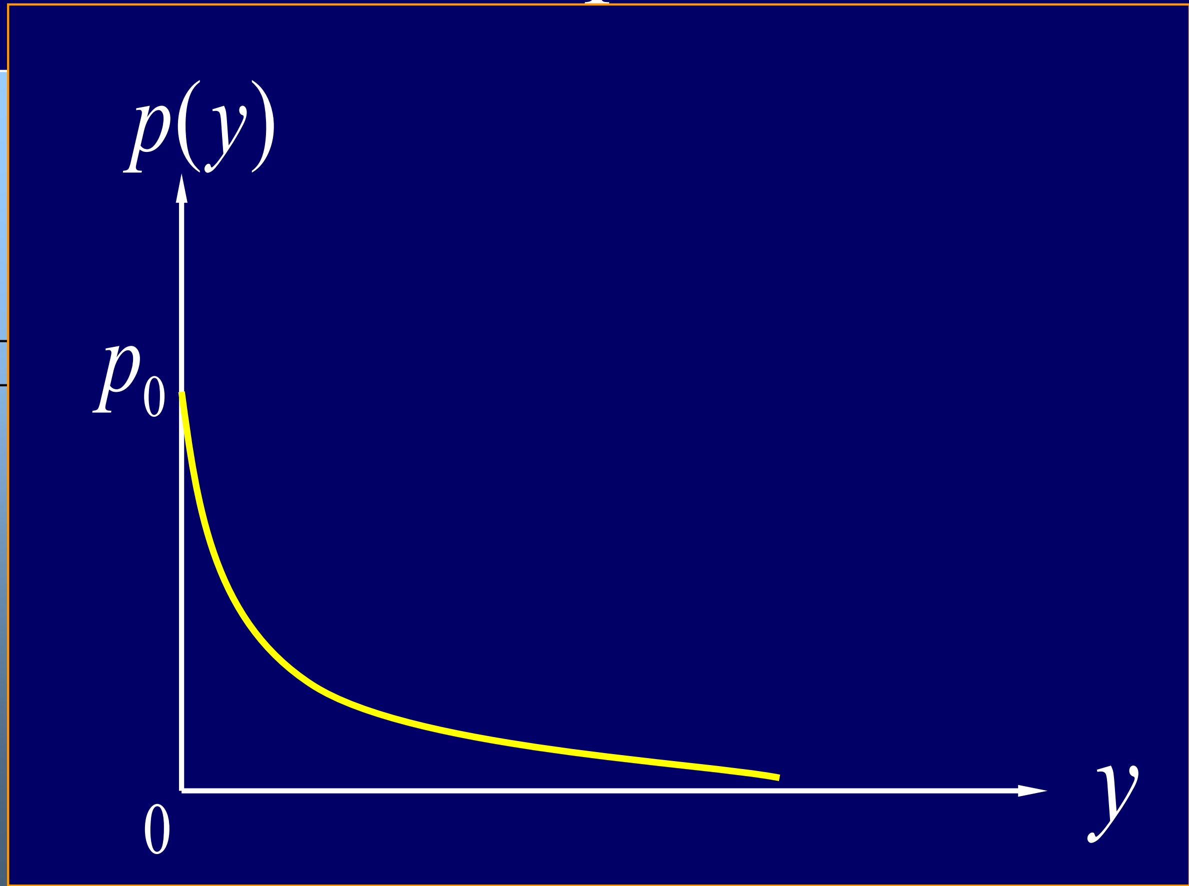
$$p(y,r) = p_0 + \rho g y + \frac{\rho\omega^2 r^2}{2}$$

Pressure in atmosphere

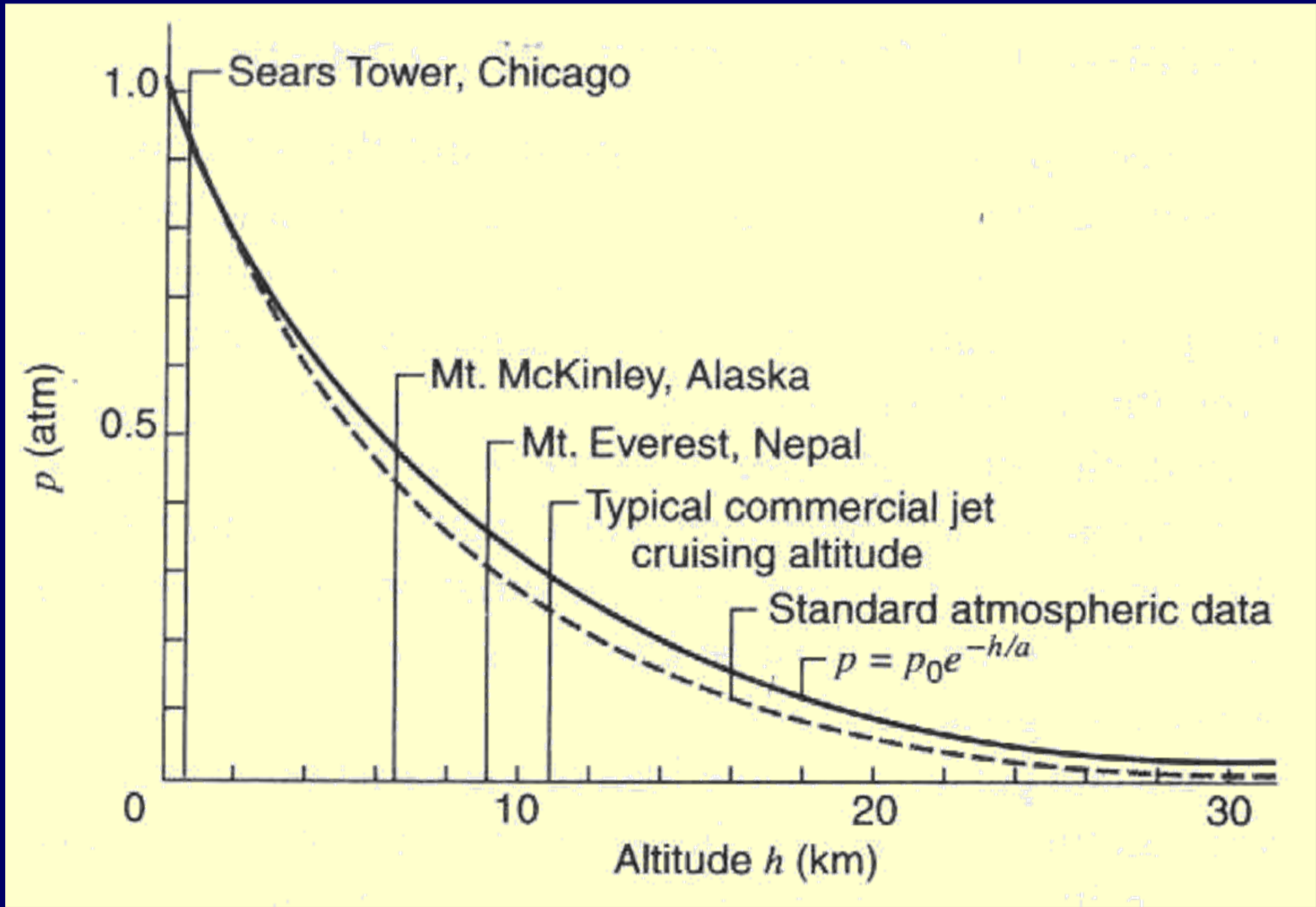
$$dp = -dy\rho g$$

$$pV = NkT$$



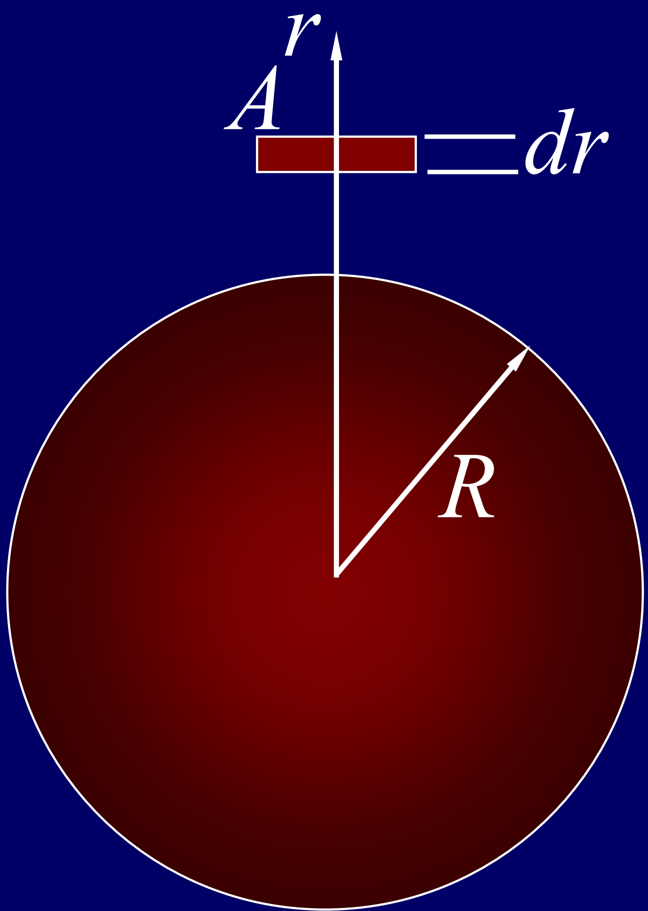


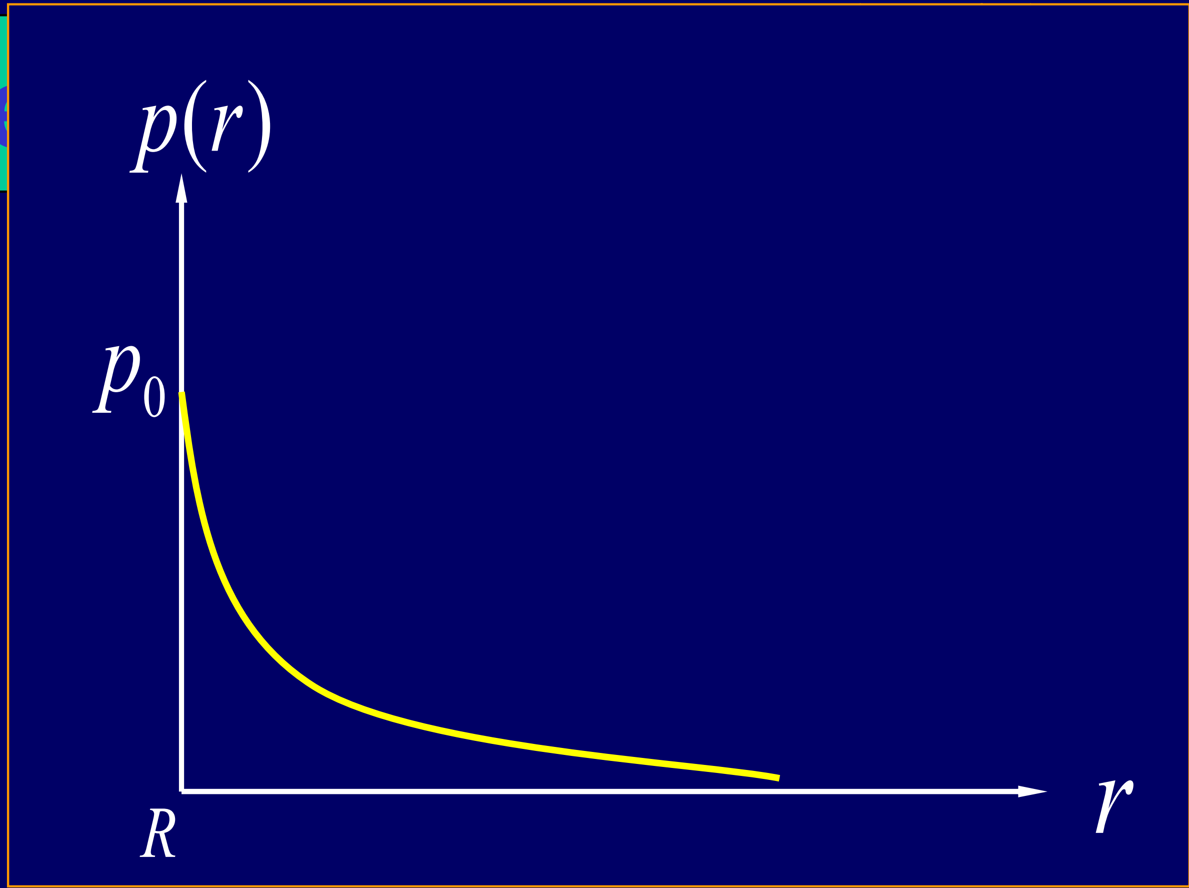
$$p = p_0 e^{-\frac{\rho_0 g y}{P_0}}$$


Sears Tower, Chicago
1.0
0.5
0
p (atm)
Mt. McKinley, Alaska
Mt. Everest, Nepal
Typical commercial jet cruising altitude
Standard atmospheric data
$p = p_0 e^{-h/a}$
10
20
30
Altitude h (km)

$$p = p_0 e^{-h/a}$$

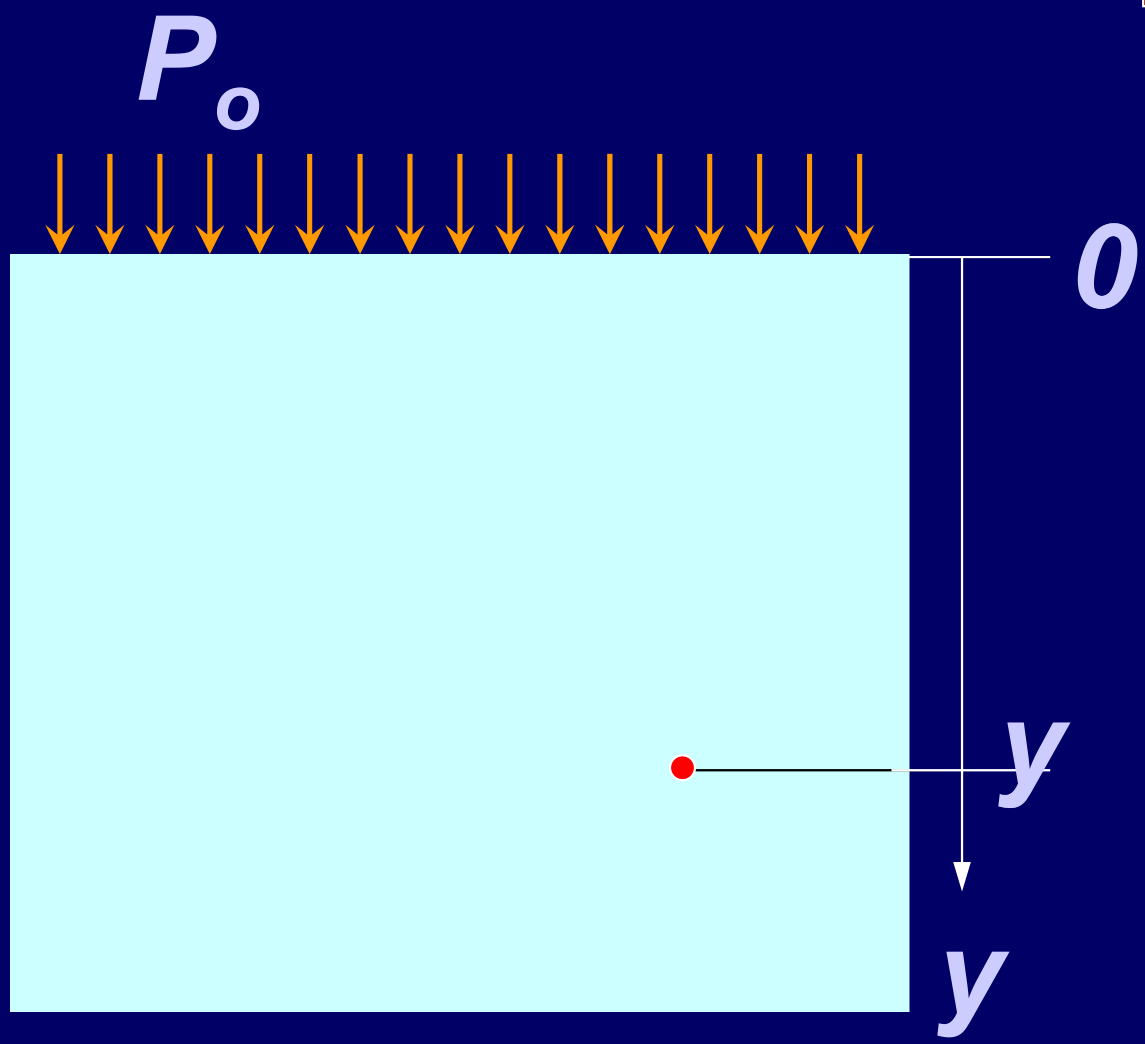
Atmosphere





$$p = p_0 e^{\frac{\rho_0 GM}{P_0}\left(\frac{1}{r}-\frac{1}{R}\right)}$$

Pressure

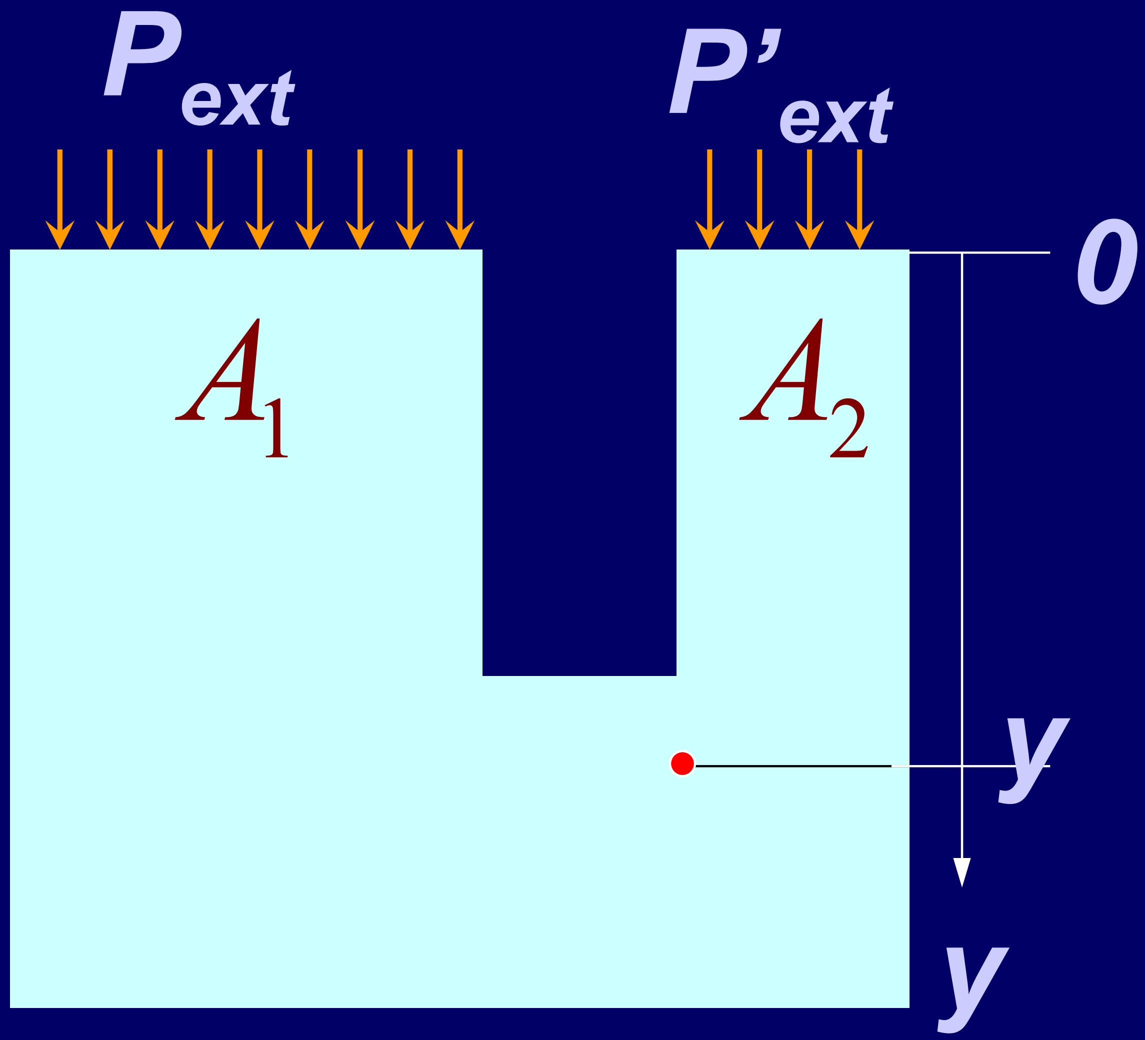


Pascal's Principle:

Pressure applied to an enclosed fluid is transmitted undiminished to every portion of the fluid and to the wall of the containing vessel.

$$P = P_o + \rho g y$$

Pressure



$$P = P_{ext} + \rho g y$$

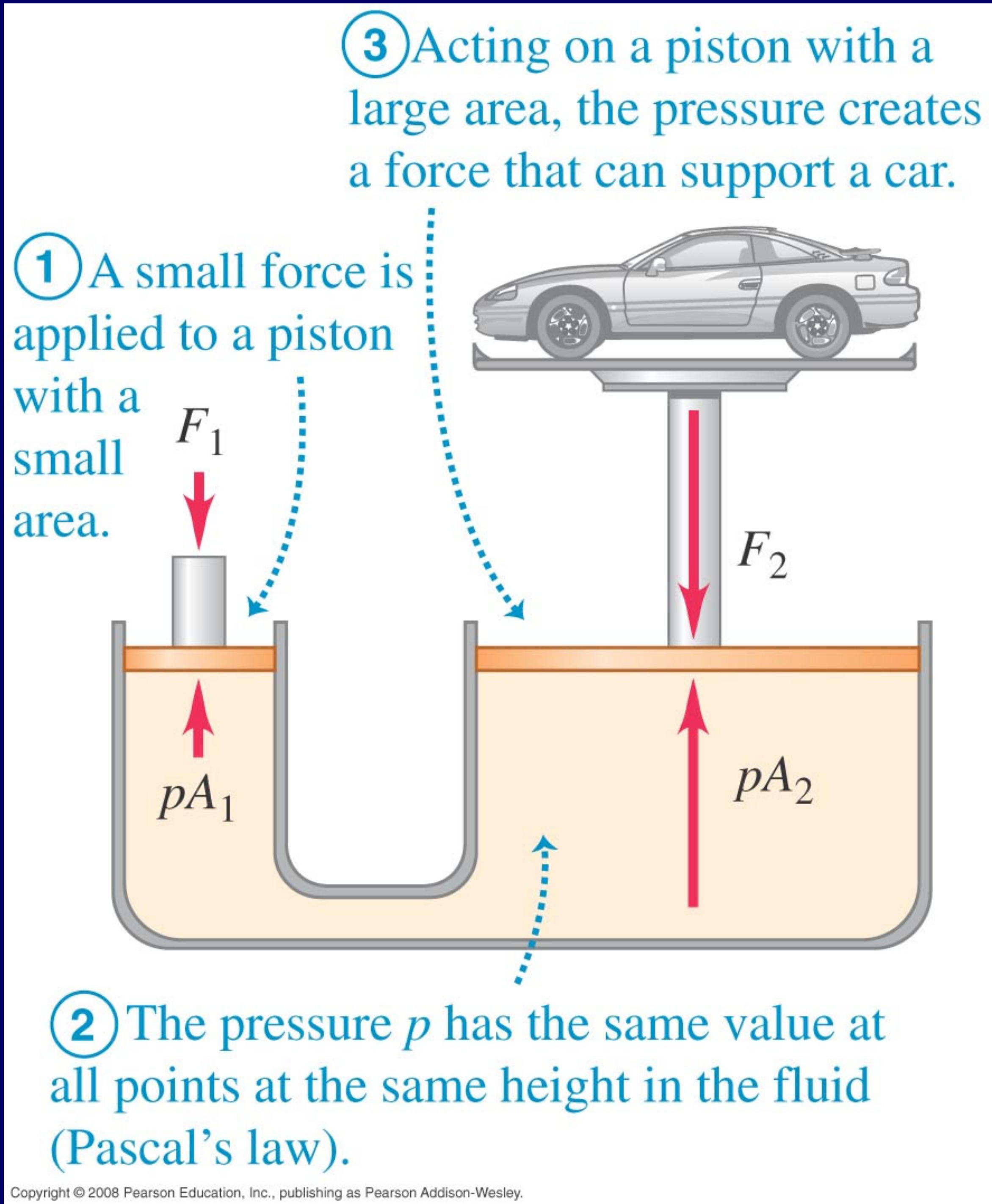
$$P = P'_{ext} + \rho g y$$

$$P_{ext} = P'_{ext}$$

$$\boxed{\frac{F_1}{A_1} = \frac{F_2}{A_2}}$$

The hydraulic lift is an application of Pascal's law. The size of the fluid-filled container is exaggerated for clarity.

$$\frac{F_i}{F_o} = \frac{A_i}{A_o}$$




LANCIA
ZACATO
ZAG

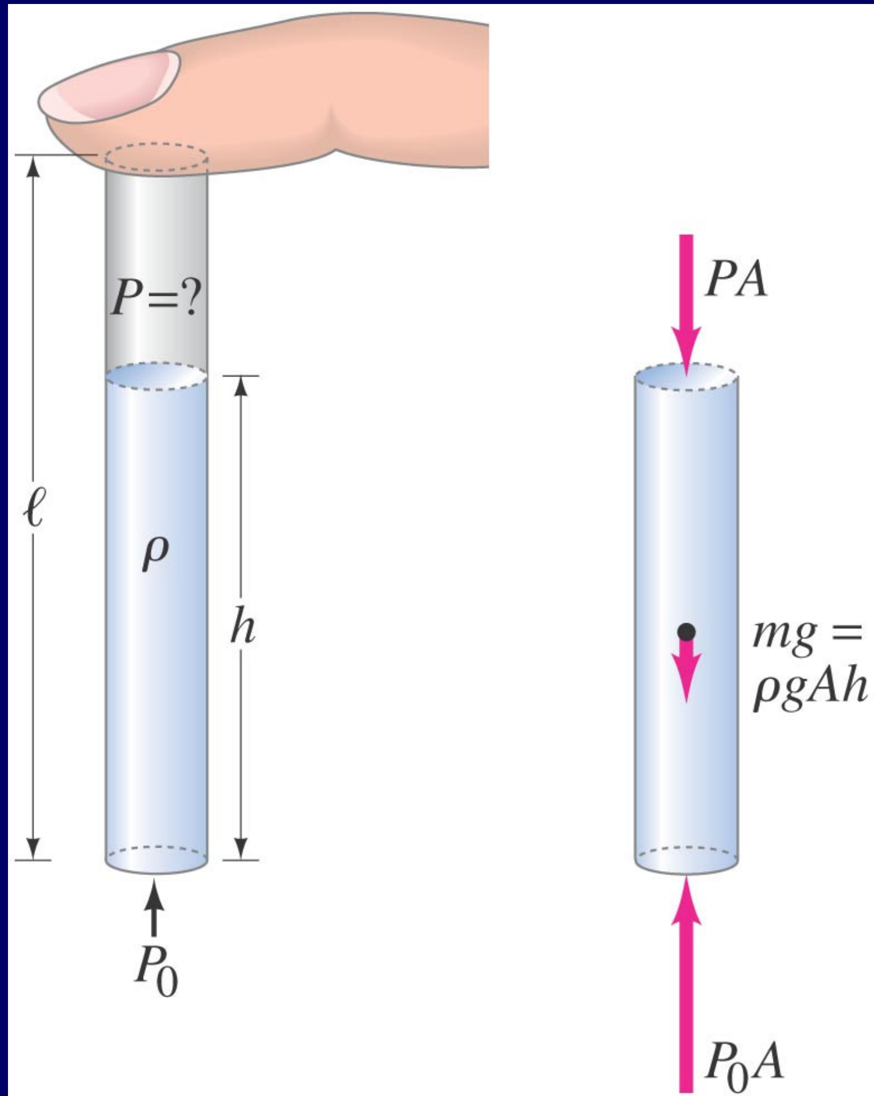
Pascal's Principle and Archimedes's Principle

You insert a straw of length l into a tall glass of water.

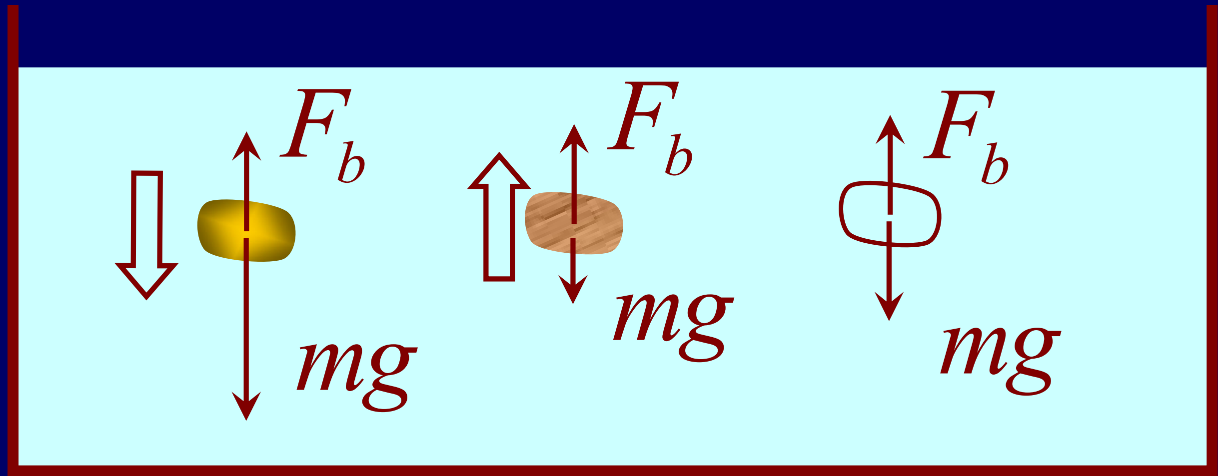
You place your finger over the top of the straw, capturing some air above the water but preventing any additional air from getting in or out, and then you lift the straw from the water.

You find that the straw retains the water.

$$\Delta p = P_0 - P = \rho g h$$



Buoyant Force $F_b = mg = \rho V g$



Archimedes's Principle:

A body wholly or partially immersed in a fluid is buoyed up by a force equal in magnitude to the weight of the fluid displaced by the body

Example: Archimedes: Is the crown gold?

Solution:

$$w' = F_T' = w - F_B$$

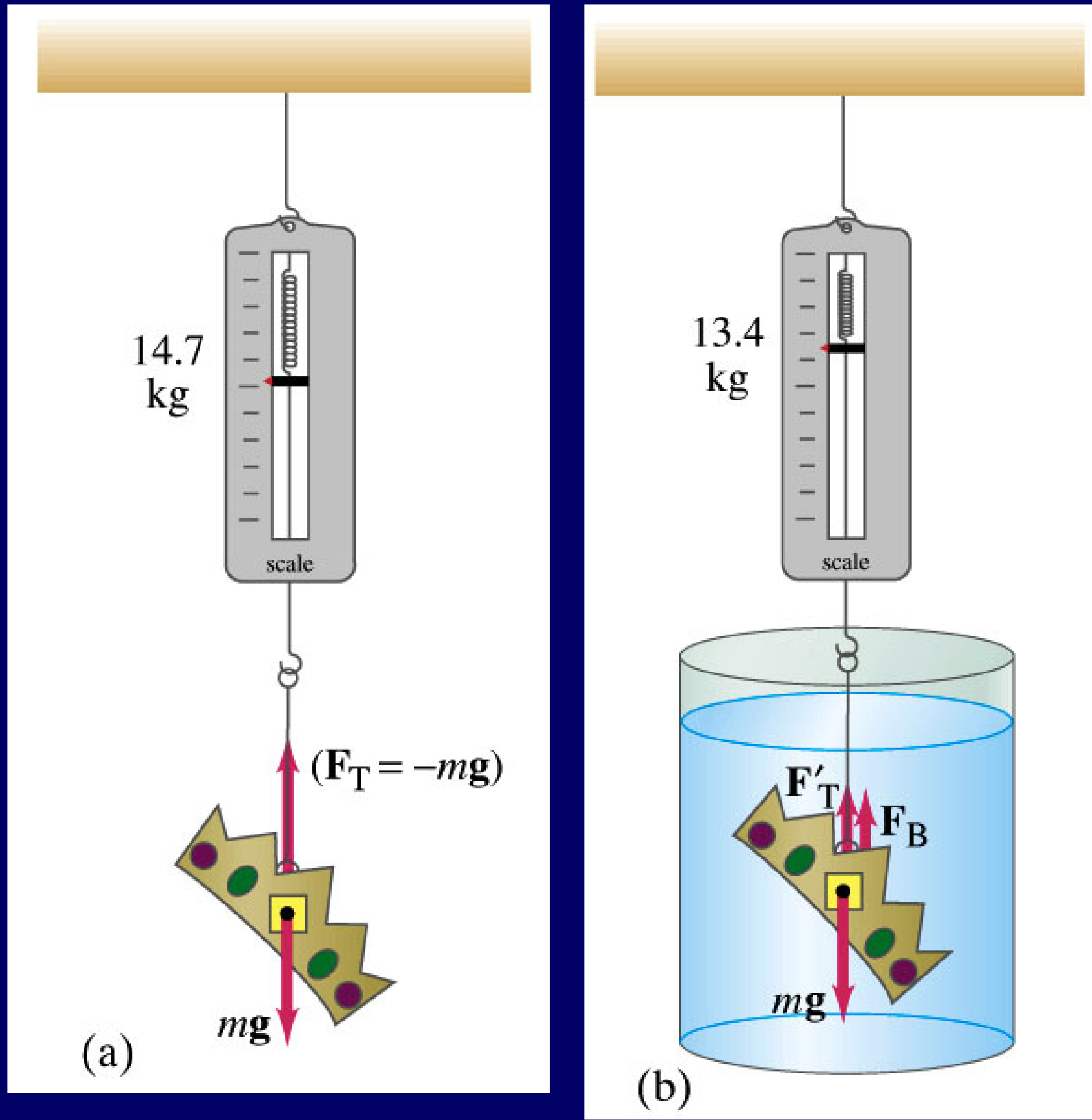
$$F_B = w - w' = \rho_F g V$$

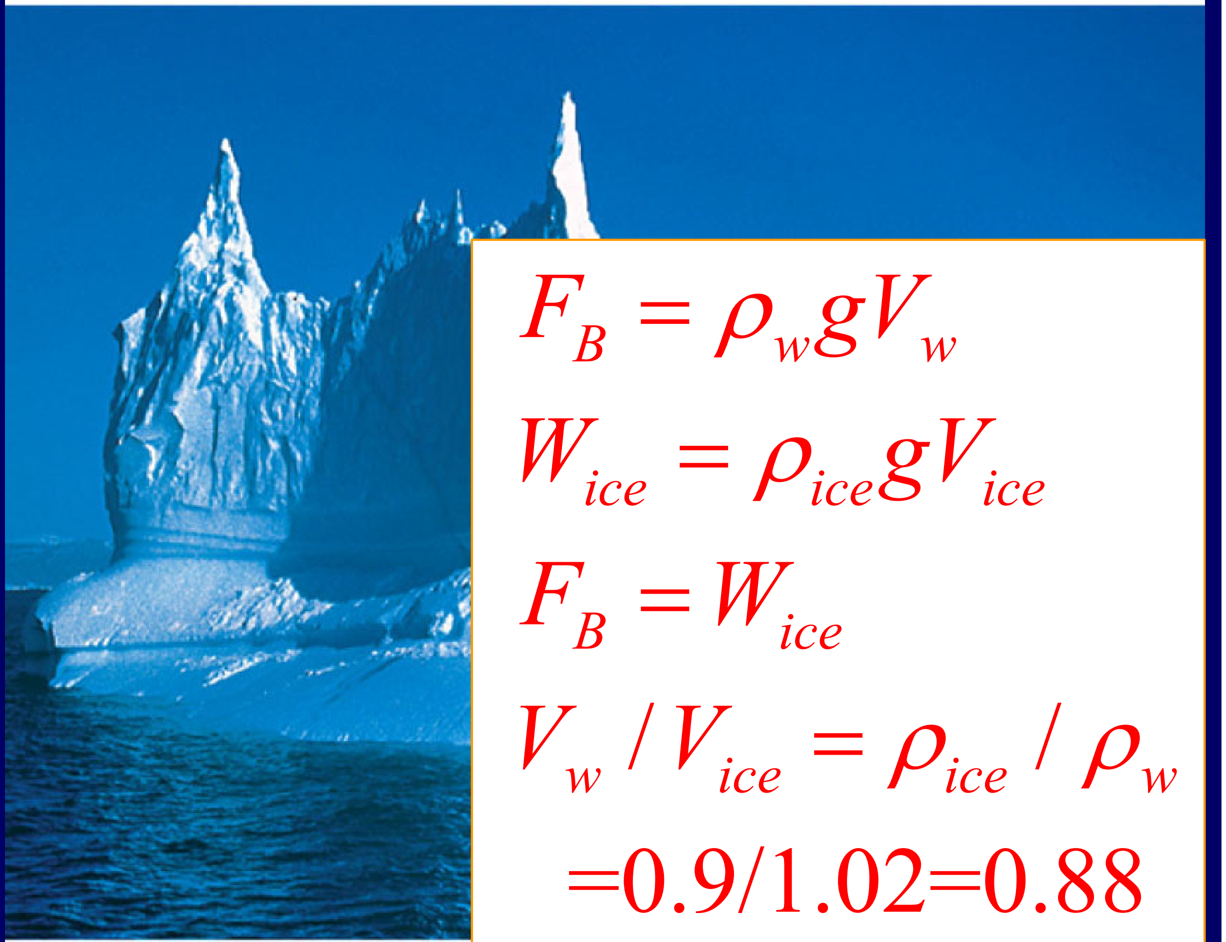
$$w = \rho_O g V$$

$$\frac{\rho_O}{\rho_F} = \frac{w}{w - w'} = 11.3$$

gold should be $19.3 g/cm^3$

$11.3 g/cm^3$ means it is lead





$$F_B = \rho_w g V_w$$

$$W_{ice} = \rho_{ice} g V_{ice}$$

$$F_B = W_{ice}$$

$$V_w / V_{ice} = \rho_{ice} / \rho_w$$

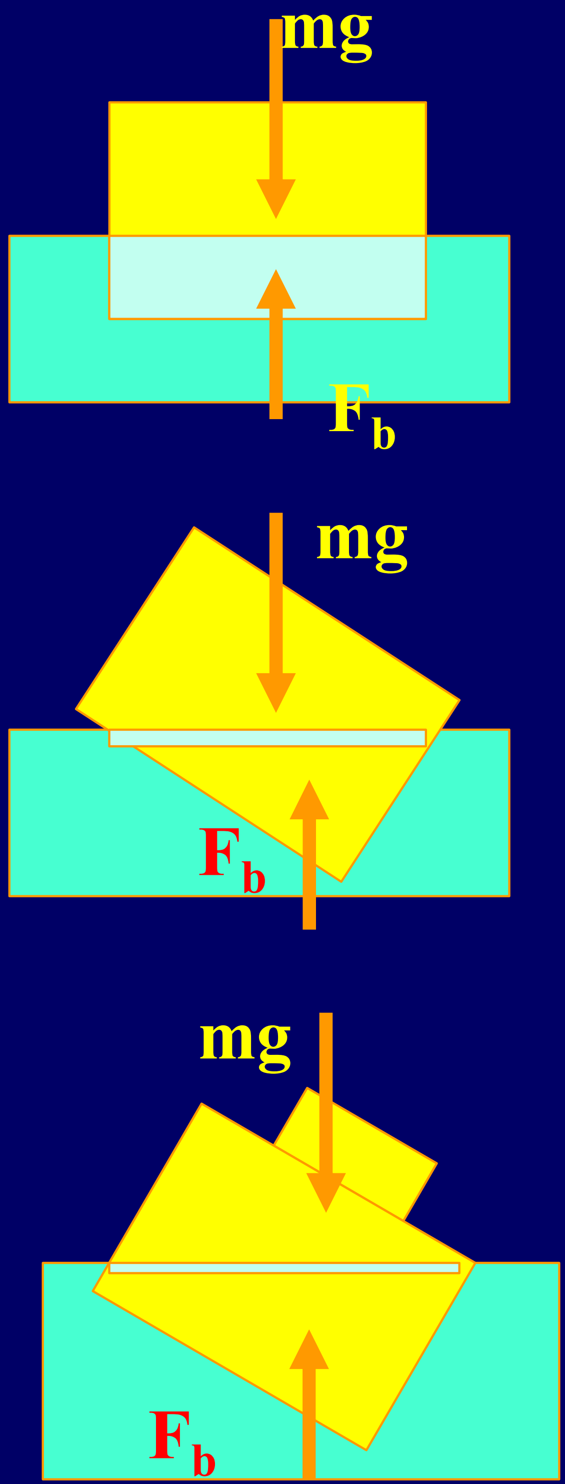
$$=0.9/1.02=0.88$$

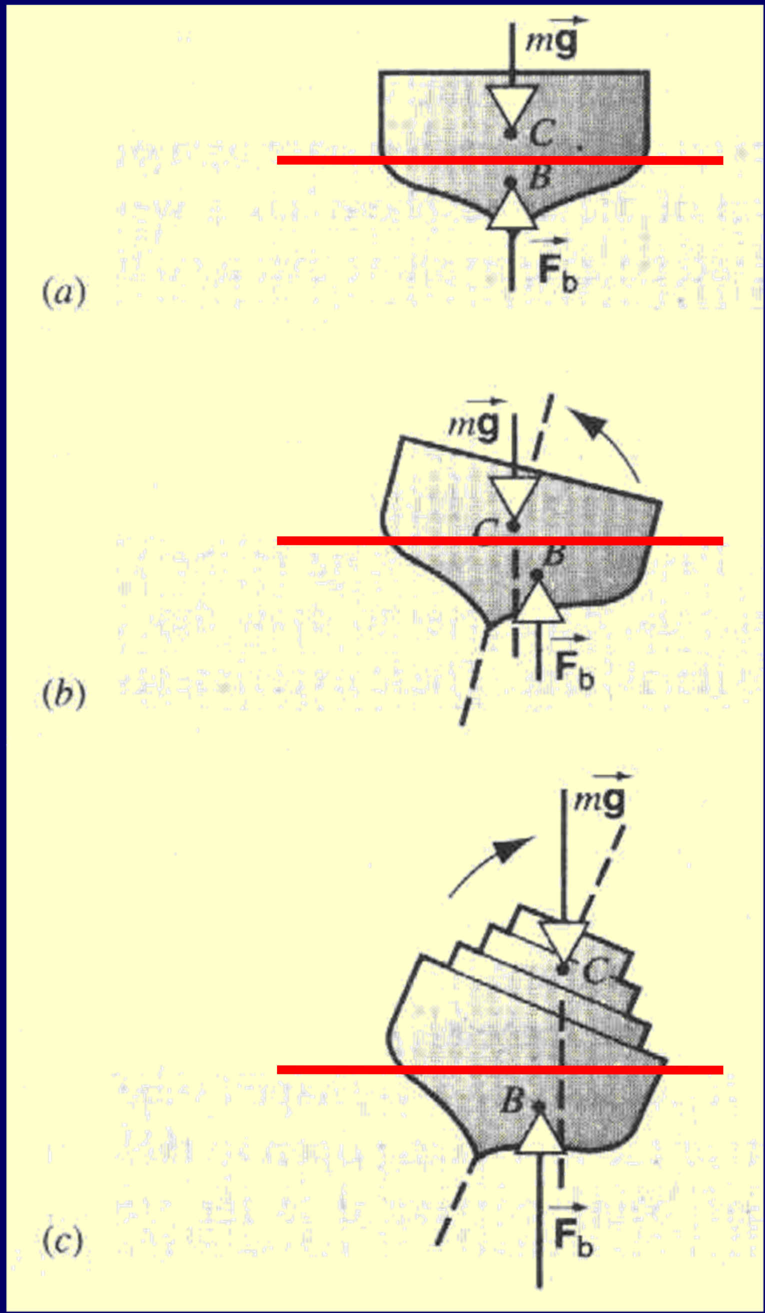
(a)

(b)

A Titanic Surprise

That's only the tip of the iceberg(12%)

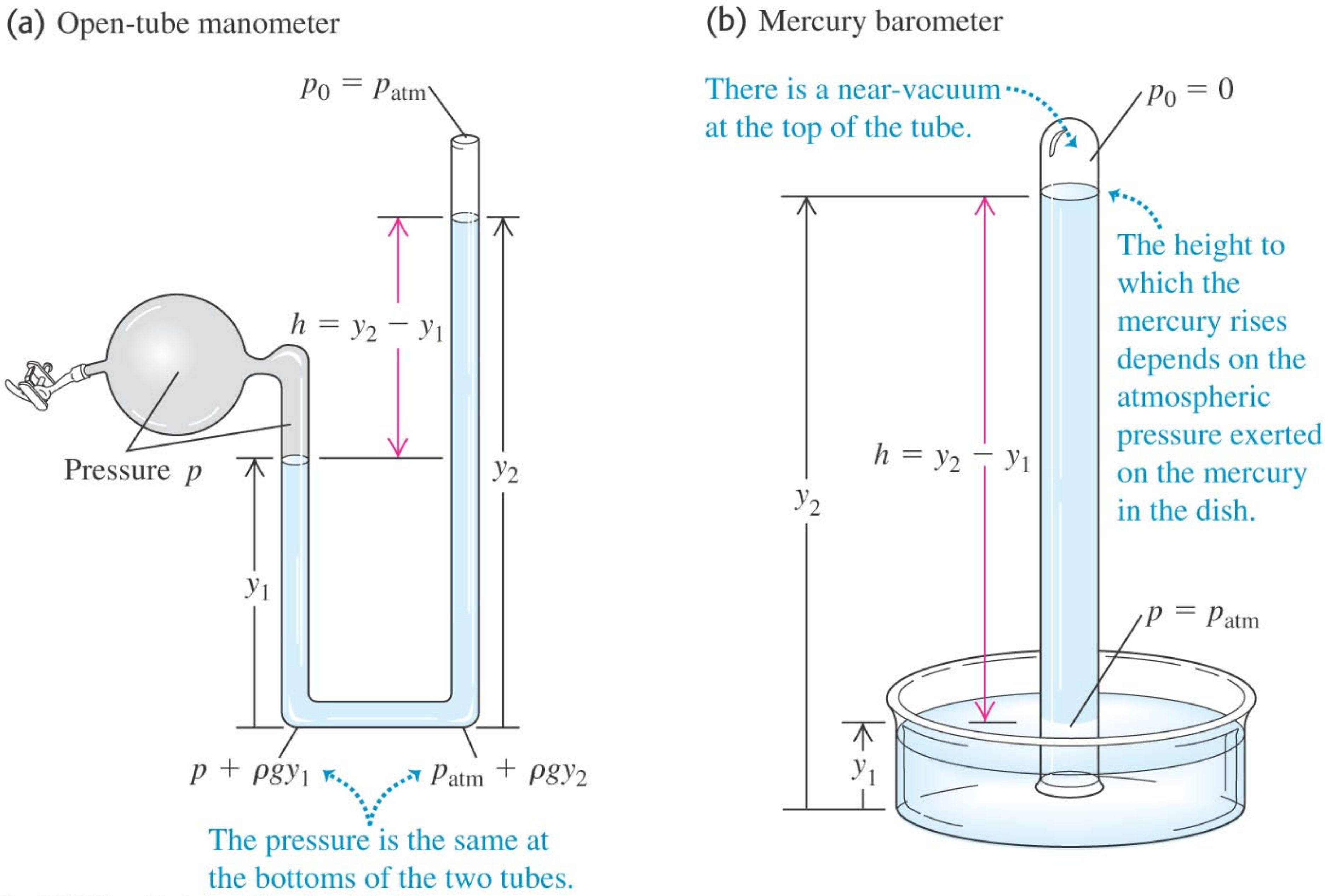




Note that the center of buoyancy

zwma@zju.edu.cn

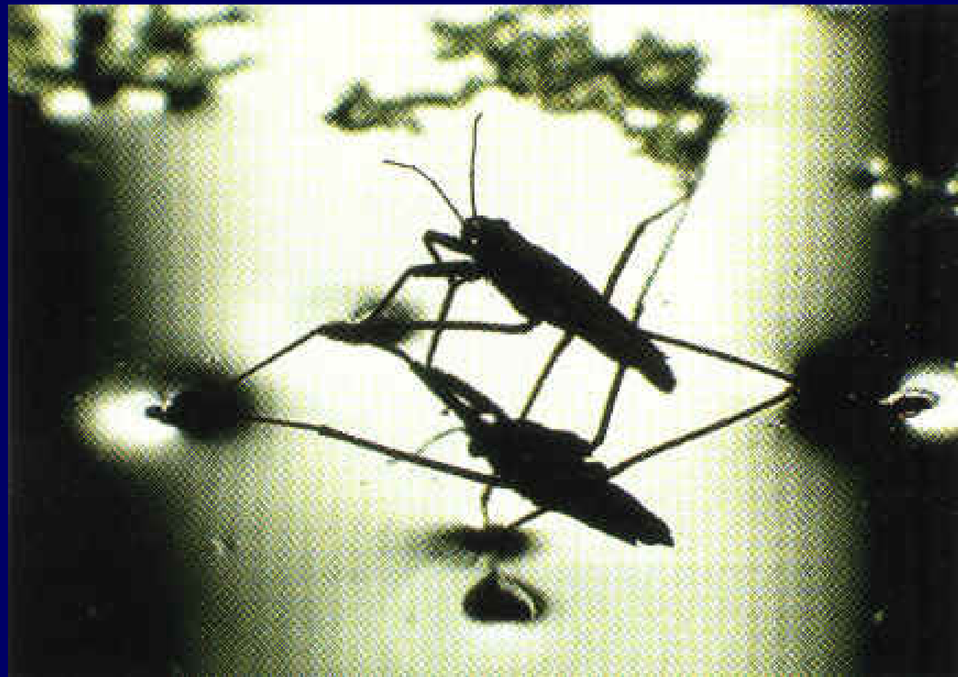
Measurement of Pressure



Surface Tension and Capillarity

Question:

Capillarity

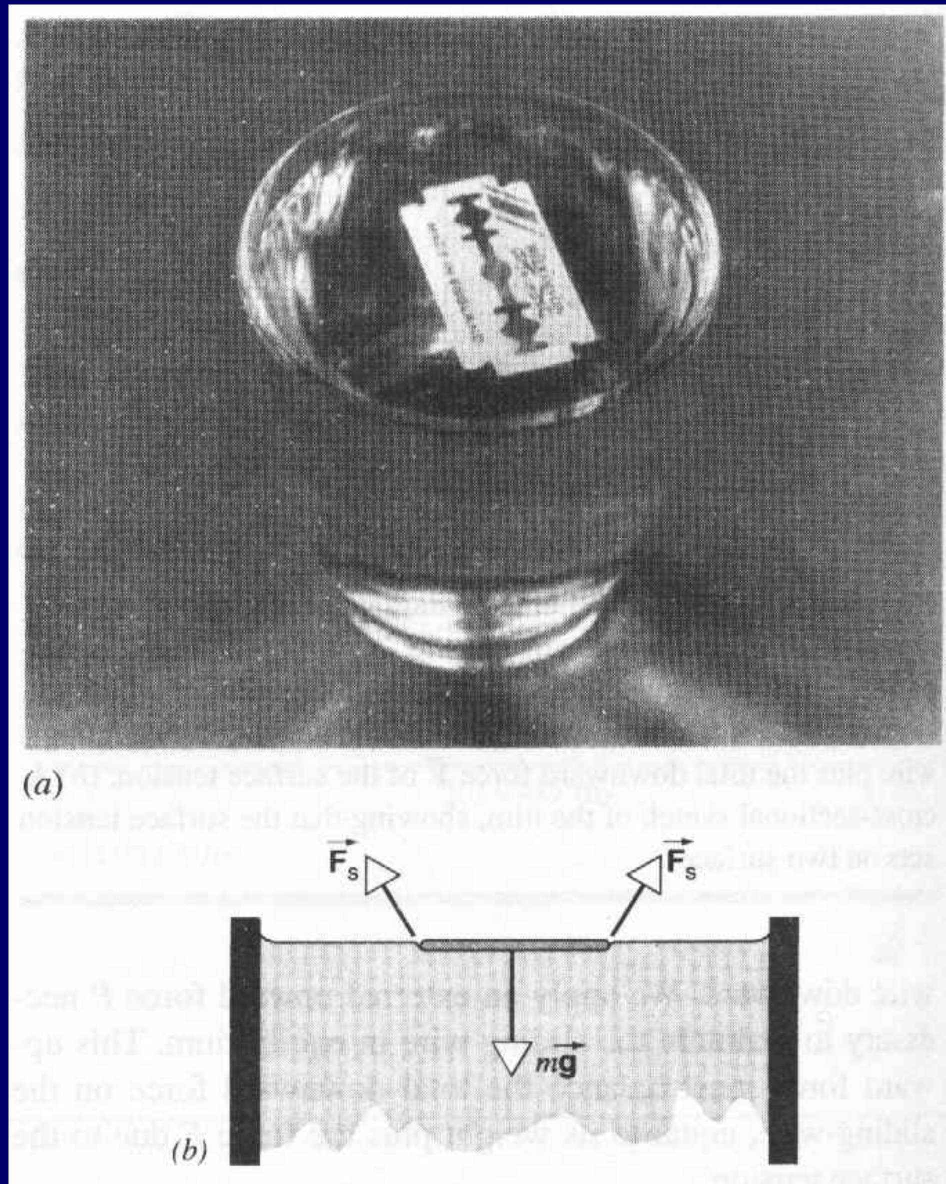


Surface Tension

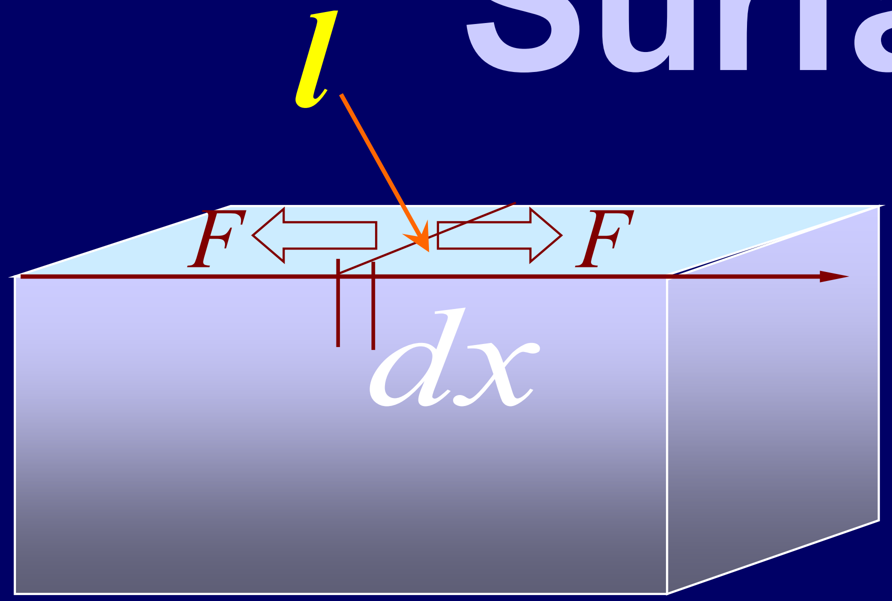
Surface tension

$$F = \gamma l \qquad \gamma = \frac{F}{l}$$

γ is the surface force per unit length.



Surface tension



$$dW = F \cdot dx$$
$$= -\gamma\, l \cdot dx$$
$$= -\gamma\, dA$$

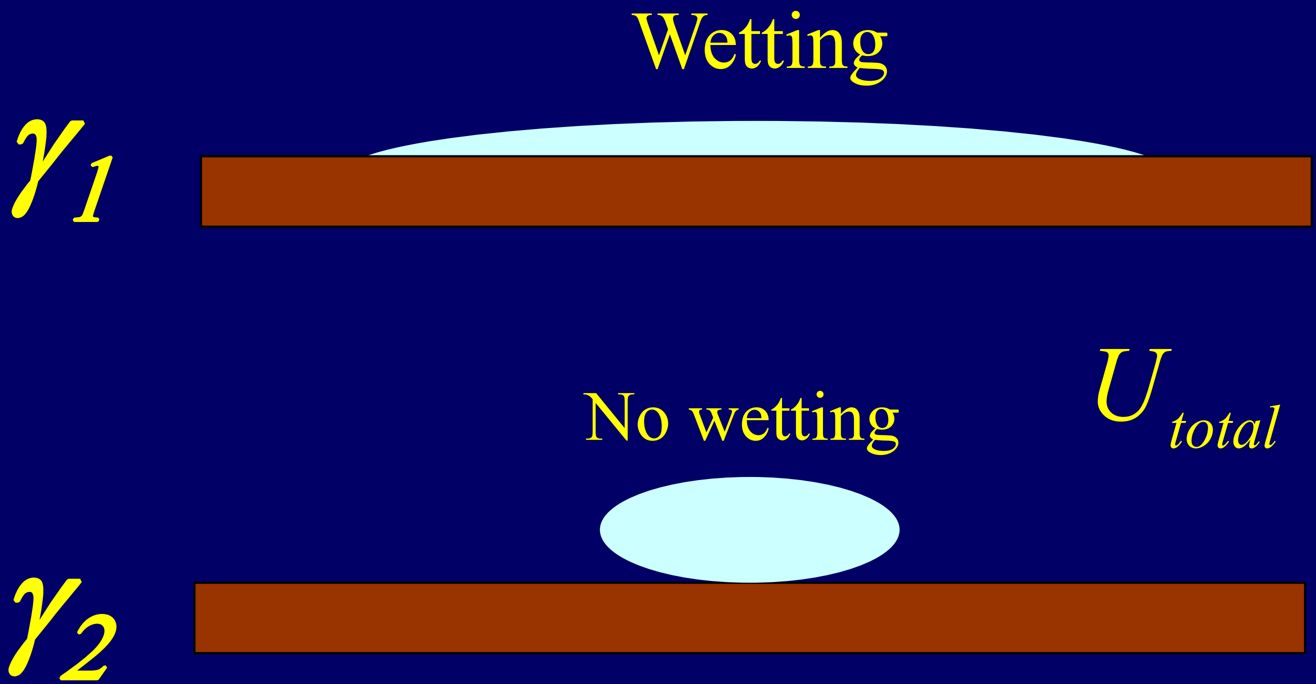
$$\boldsymbol{F = \gamma\, l}$$

$$dU = -dW = \gamma\, dA$$

Surface tension potential energy

$$\gamma = \frac{F}{l}$$

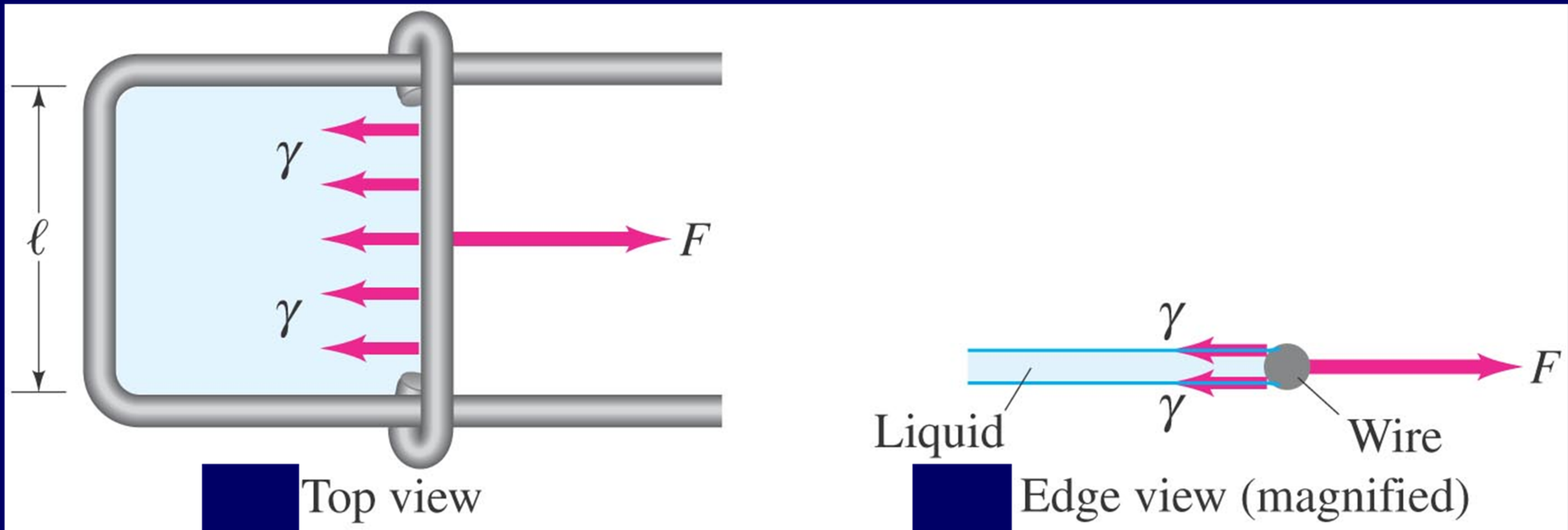
$$\gamma = \frac{dU}{dA}$$



$$\gamma = \frac{dU}{dA}$$

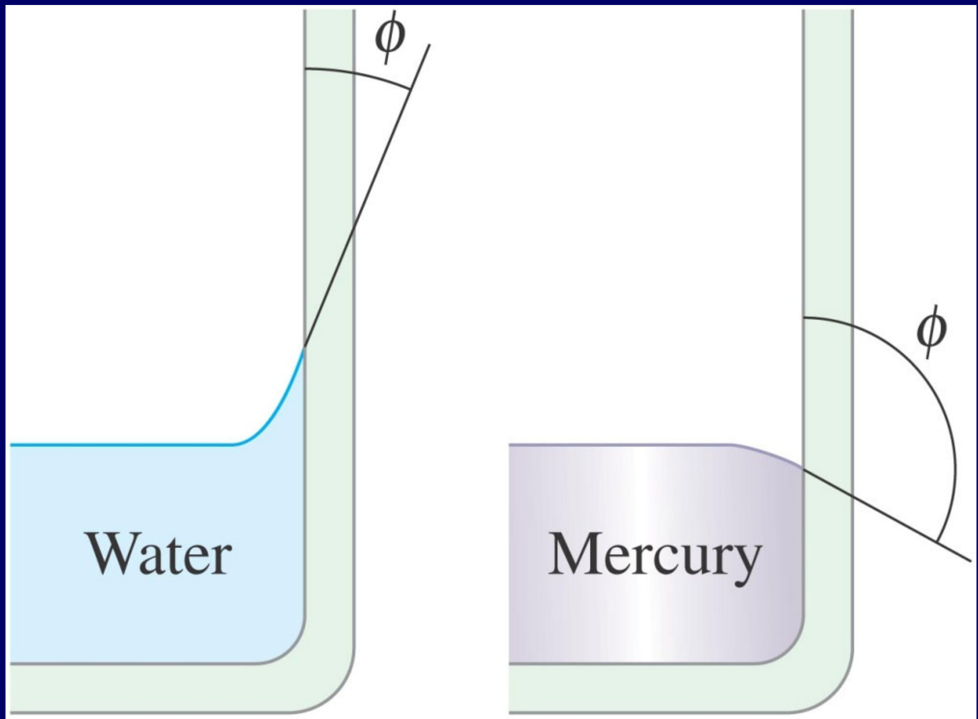
$$U_{total} = U_{surface} + U_{grav.}$$

$$U_{t1} = U_{t2} \quad U_{g1} \approx U_{g2} \qquad \gamma_1 < \gamma_2$$


γ
γ
ℓ
F
Top view
γ
γ
F
Liquid
Wire
Edge view (magnified)

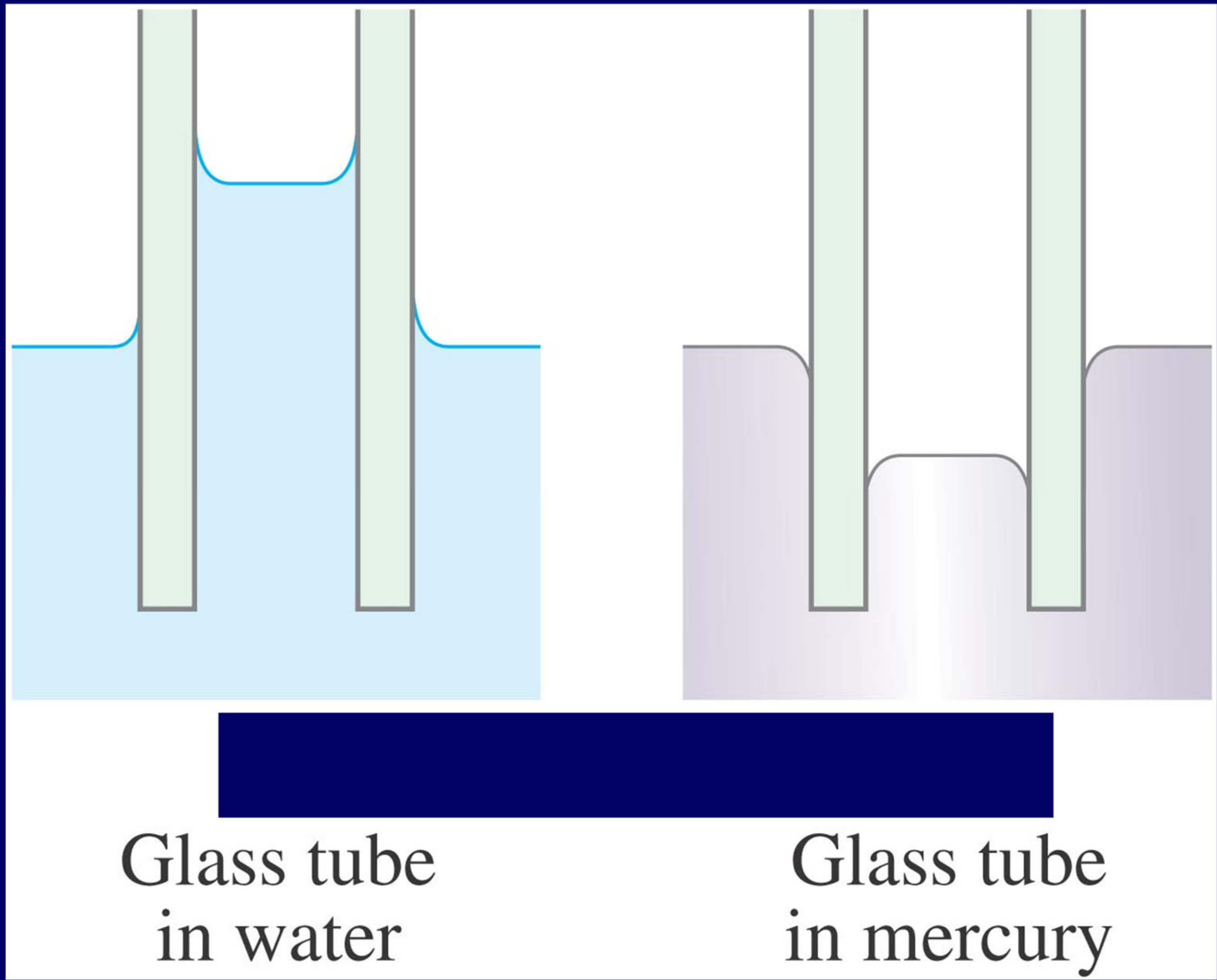
Because of surface tension, some objects more dense than water may not sink.

Soap and detergents lower the surface tension of water. This allows the water to penetrate materials more easily.



Water molecules are more strongly attracted to glass than they are to each other; just the opposite is true for mercury.

If a narrow tube is placed in a fluid, the fluid will exhibit capillarity.



CHAP 15

Problems

P348:7, 8,11,13,17